Oops AI Project Proposal

Automated KPI Root Cause Analysis Engine

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Problem Statement and Target Users

Modern organizations track dozens or hundreds of business metrics such as revenue, conversion rates, churn, customer satisfaction, and operational KPIs. When a metric suddenly drops or spikes, analyses and managers typically manually investigate dashboards, query multiple data sources, and test hypotheses to identify the root cause. This process is time consuming and repetitive, error-prone, and difficult for non-technical stakeholders to interpret. As a result, organizations often respond slowly to critical issues or make decisions based on incomplete explanations.

Target Users

- Data Analysts/ Business Analysts - automate first-pass root cause analysis
- Product Managers - understand metric changes
- Operations - diagnose performance issues
- Executives/ Non-technical stakeholders - receive clear natural-language explanations

Proposed Solution and Key Features

The system functions as an “AI metric detective” that turns raw data into actionable insight.
We propose building an LLM-powered system that automatically:

- Detects significant changes (anomalies) in selected KPIs
- Analyzes related features and segments in the data
- Identifies and ranks likely root causes using statistical methods
- Generates a human-readable explanation report using a large language model

Key Features:

1. KPI monitoring and anomaly detection
    a. time-series monitoring of selected metrics
    b. detection using statistical thresholds or ML-based anomaly detection
2. Automated Root Cause Analysis
    a. compare metric behavior across dimensions (e.g., region, product, user type, etc.)
    b. statistical testing (e.g., correlation, regression, hypothesis tests)
    c. ranking causes by effect size and confidence
3. LLM Explanation
    a. converts statistical findings into structured natural language reports
    b. explains what changed, why it likely happened, and what to check next
4. Interactive Dashboard
    a. view metrics, view anomaly alerts, view ranked causes and explanations

Technical Stack

Component	Technology
Frontend	Streamlit
Data Processing	Python: pandas, numpy
Statistical Engine	Statsmodels, scipy.stats
LLM API	Claude API
LLM Framework	LangChain
Visualization	

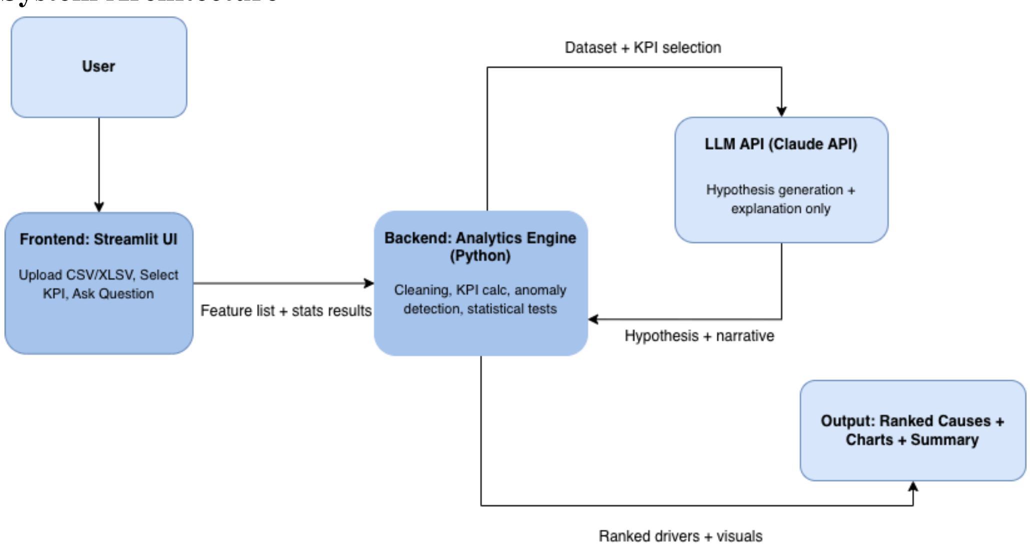
Frontend
- Streamlit will be the user interface for data upload, KPI selection, and result visualization

Backend
- Python
- Pandas/NumPy for data processing
- SciPy and statsmodels for statistical tests
- Scikit-learn for optional anomaly detection

LLM
- Claude API will be used for hypothesis generation and executive level explanations

Data Input
- CSV and/or excel files uploaded by users
- SQL?

System Architecture



Success Criteria and Evaluation Metrics

Functional Success:

- System correctly identified known root causes in synthetic test datasets
- LLM hypotheses are 100% constrained to actual dataset columns
- End-to-end analysis completes in under 60 seconds for a typical dataset

Academic Alignment:

- Demonstrates statistical reasoning, hypothesis testing, and feature analysis
- Shows responsible LLM usage with grounded, explainable outputs
- Delivers business ready storytelling through executive summaries